Ahmed Ikram | AI Engineer

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Final-year BSc Computer Science (Data Science & AI), Dundee (predicted First, June 2027). Built agentic-RAG, LLMOps and ML systems in Agile delivery, including an NCR Atleos industry project with weekly stakeholder demos - translating progress for an engineer and a non-technical PM, shaping the build on feedback. Self-taught Terraform to production standard. Mentored junior pupils one-to-one. Seeking a graduate AI Engineering role from June 2027.

EDUCATION

University of Dundee - *BSc (Hons) Computer Science (Data Science and AI)* *Sep 2023 - Present*

- On track for a First-Class Honours degree. Expected Graduation: June 2027
- Honours Project: Tiny Language Model Laboratory - Transformer Learning-Dynamics Research Platform

  Building a configurable PyTorch Transformer laboratory to investigate how small language models learn from synthetic algorithmic tasks. Training and evaluating models across changes in capacity, dataset size, task difficulty and regularisation, with experiments focused on memorisation, overfitting, generalisation and the potential emergence of grokking.

High School of Dundee *Sep 2017 - June 2023*

- Highers: A Mathematics, A Computing Science, A English, B Chemistry. Advanced Highers: Computing Science C.
- Computing Science classroom assistant - mentored junior pupils one-to-one during classwork.

SKILLS

Languages & Data: Python · SQL · TypeScript · JavaScript · Java · C · Bash · Rust / PyO3 · PostgreSQL · Redis · Kafka · Pydantic

LLMs: PyTorch · Hugging Face Transformers · QLoRA/LoRA/PEFT · AWQ · Unsloth · TRL · Accelerate · BitsandBytes · vLLM · Jinja2, Prompt & Context Engineering · TypeSafe System One (Jev) · OpenRouter (LLM Gateway)

Agents & RAG: LangGraph · LangChain · Model Context Protocol (MCP) · AWS Bedrock · ChromaDB · Cross-Encoder Reranking · Multimodal (Vision-Language) Extraction · Guardrails

Evaluation & Observability: RAGAS · Calibration (ECE) · LLM-as-Judge · LangSmith · Langfuse · Evidently AI · OpenTelemetry (Jaeger)

MLOps & Serving: MLflow · Weights & Biases · Apache Airflow · FastAPI · GraphQL · Modal · AWS SageMaker

Classic ML: XGBoost · Isolation Forest · scikit-learn · Optuna · Tesseract · OpenCV

Cloud & DevOps: AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, WAF, IAM, CloudFront, Secrets Manager, DynamoDB) · Google Cloud (GCS) · Terraform · Docker · GitHub Actions / CI/CD · Linux · Dependabot · CodeQL

PROJECTS

ATM Log Aggregation, Anomaly Detection & Diagnostics - RAG over Logs and ML Anomaly Detection (NCR Atleos)

LangGraph · LangChain · MCP · ChromaDB · FastAPI · XGBoost · MLflow · Kafka · Redis · Docker · Terraform · AWS

- Built a Hybrid RAG LangGraph agent over a 12-tool MCP server: hybrid retrieval (dense + BM25 → RRF k=60), cross-encoder reranking, grounding-gated retry & a System One (TypeSafe Jev) decision layer for typed confidence/critique/intent decisions, 18x cheaper; custom OpenTelemetry rag.* & decision.* spans to track latency, token usage, and per-decision cost.
- Evaluated 3 retrieval strategies (baseline, hybrid-only, agentic) with RAGAS on a 50-query human-verified golden set: agentic recall 0.708 vs hybrid-only 0.350 with the best precision of 0.874; calibrated the decision layer against human verdicts, agreeing 83% of judgements, with an empirically tuned escalation threshold; CI blocks RAGAS regression >0.05.
- Architected 3-layer anomaly detection pipeline (XGBoost-8-class + Isolation Forest, Z-score drift, 7-heuristics) on live KRaft ATM telemetry streams; achieved 92% macro-F1 on held-out data and 94% on a temporal holdout - champion-versioning in MLflow with AWS Sagemaker cross-checking.
- Terraform-provisioned AWS (ECS Fargate, RDS, S3, SageMaker, Secrets Manager, 4 least-privilege IAM roles, state locked in DynamoDB) behind OIDC CI/CD, validated by 1,974 tests across pytest, Vitest, Playwright and Terraform assertions.

StockLens FinTech Spending & Investment App - AI Agent, Multimodal Extraction, MLOps

LangGraph · MCP · PyTorch · Optuna · Rust / PyO3 · Airflow · MLflow · Evidently AI · Tesseract · GraphQL · Redis · Terraform

- Built a 2-node ReAct StateGraph agent exposing 16 tools to multi-provider (AWS Bedrock) via a Stateless MCP server (Streamable HTTP, OAuth 2.1 PKCE). Correctness judged in CI by an LLM-as-judge with LangSmith tracing.
- Streamed agent responses via an SSE endpoint: reasoning blocks and tool-call steps as distinct events before the final response - with tiered conversation memory (Redis hot cache, Postgres cold archive) and turn-budget-gated compaction.
- Shipped a confidence-gated receipt-extraction cascade (Tesseract OCR → Bedrock Vision-LLM fallback) that only spends LLM budget when OCR confidence drops, matched to merchant with rapidfuzz.
- Trained a 5-seed LSTM ensemble (embeddings, focal loss, Rust/PyO3 features) to 53.2% 5-day directional accuracy vs 50% chance baseline and +2.3pp over GBM, with leakage controls (10-day purge embargo and train-only normalisation).
- Built real-return PnL backtests (non-overlapping 5-day equal-weight portfolios, annualised, 0/10bps costs): long-only Sharpe ~2.9 net of costs, long-short ≈0 - both reported.
- Automated weekly Airflow retraining: 30-trial Optuna HPO with a paired champion/challenger promotion gate - >2pp gain with exact McNemar $p<0.05$ against the re-scored champion; Evidently drift monitoring (PSI/KS/JSD) before production.
- Shipped on a fully Terraform-provisioned AWS estate (ECS Fargate, S3-delivered champion) through 9 parallel CI jobs and an 8-job, 5-stage OIDC CD with 2,417 tests at a 90% coverage gate.

SWE-Qwen Automated LLM Fine-Tuning, Evaluation & Serving

Python · PyTorch · QLoRA · Unsloth · Hugging Face · Modal · vLLM · Weights & Biases · Langfuse · Terraform · Google Cloud

- Architected a model-agnostic LLMOps platform: one CLI command onboards any open-weight model into the pipeline - data preparation → fine-tuning → evaluation → statistically-gated promotion → OpenAI-compatible serving; versioned GCS/W&B artifacts, with per-generation traces in Langfuse.
- Engineered a 9-stage schema-validated data pipeline processing 20,477 raw SWE-bench issues across 46 repos into 17,456 cleaned examples; prevented cross-split repository leakage and versioned in W&B and GCS.
- Fine-tuned three Qwen3-14B QLoRA variants; champion hit 7× base fail-to-pass (17.20% vs 2.46%) on a 100-instance held-out sample (95% CI 11.1-25.8%, McNemar $p\approx 6e\text{-}05$ and 90.10% unrelated pass-pass rate, +61.6 pt vs base).
- Built an execution-based evaluation harness running real regression tests in official SWE-bench Docker images; enforced a 4-condition statistical gate using confidence intervals and paired significance tests.
- Delivered scale-to-zero Async FastAPI & vLLM serving per-request LoRA switching and W&B/Langfuse observability; provisioned GCP infrastructure via Terraform & Workload Identity Federation; a 20-instance F2P smoke gate blocks regressing PRs and 1,753 tests.

CERTIFICATES

AWS Academy Graduate - Machine Learning Foundations	F*eb 2025*
Foundations of Azure AI: Concepts, Capabilities, and Implementation	*Oct 2024*
AWS Academy Graduate - Cloud Foundations	*Mar 2024*

EXPERIENCE

Beauly Mini Market, Dundee, Scotland

Retail Assistant — *Sep 2023 - Present*

Zulus Peri Peri Grill, Dundee, Scotland

Front of House Worker — *June 2021 - Sep 2023*